Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Computer
Data Privacy and Security

Data Science and Management (DASMA)

LUISS

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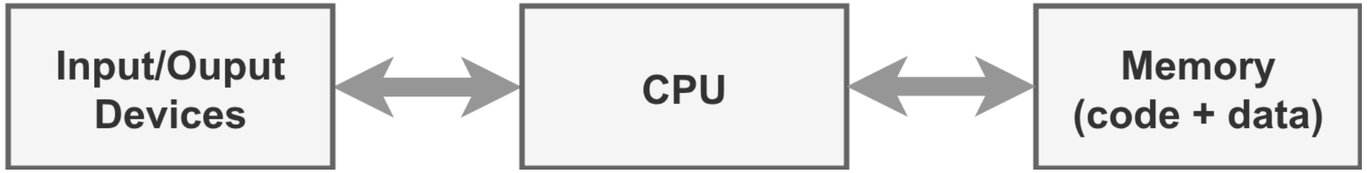
To deploy and manage these systems at scale, we may use:

4. **Virtualization**: creating multiple virtual machines on physical hardware
5. **Containerization**: packaging applications with their dependencies

Hardware

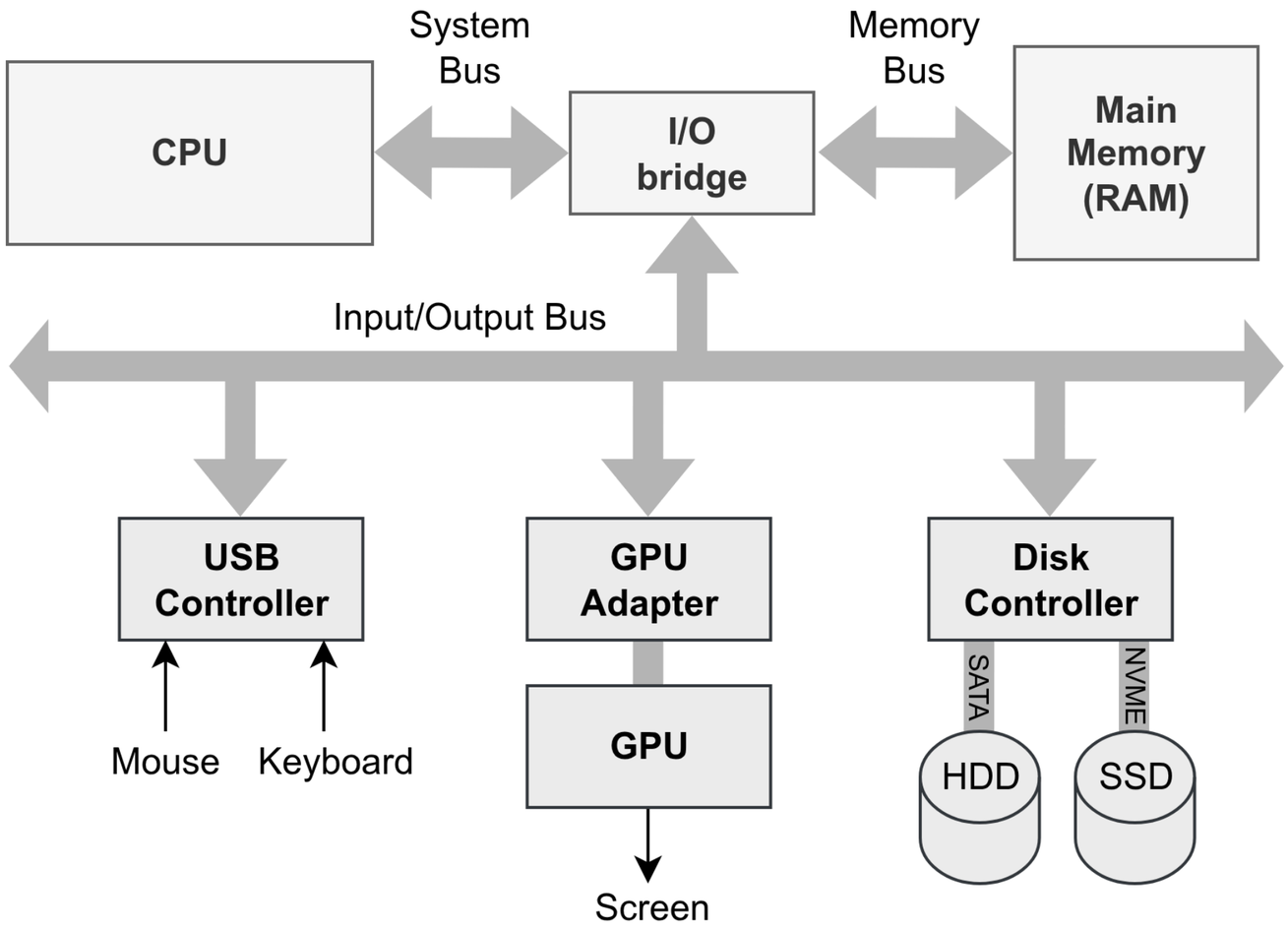
What is inside a Computer?

A (still accurate) high-level overview of a computer:

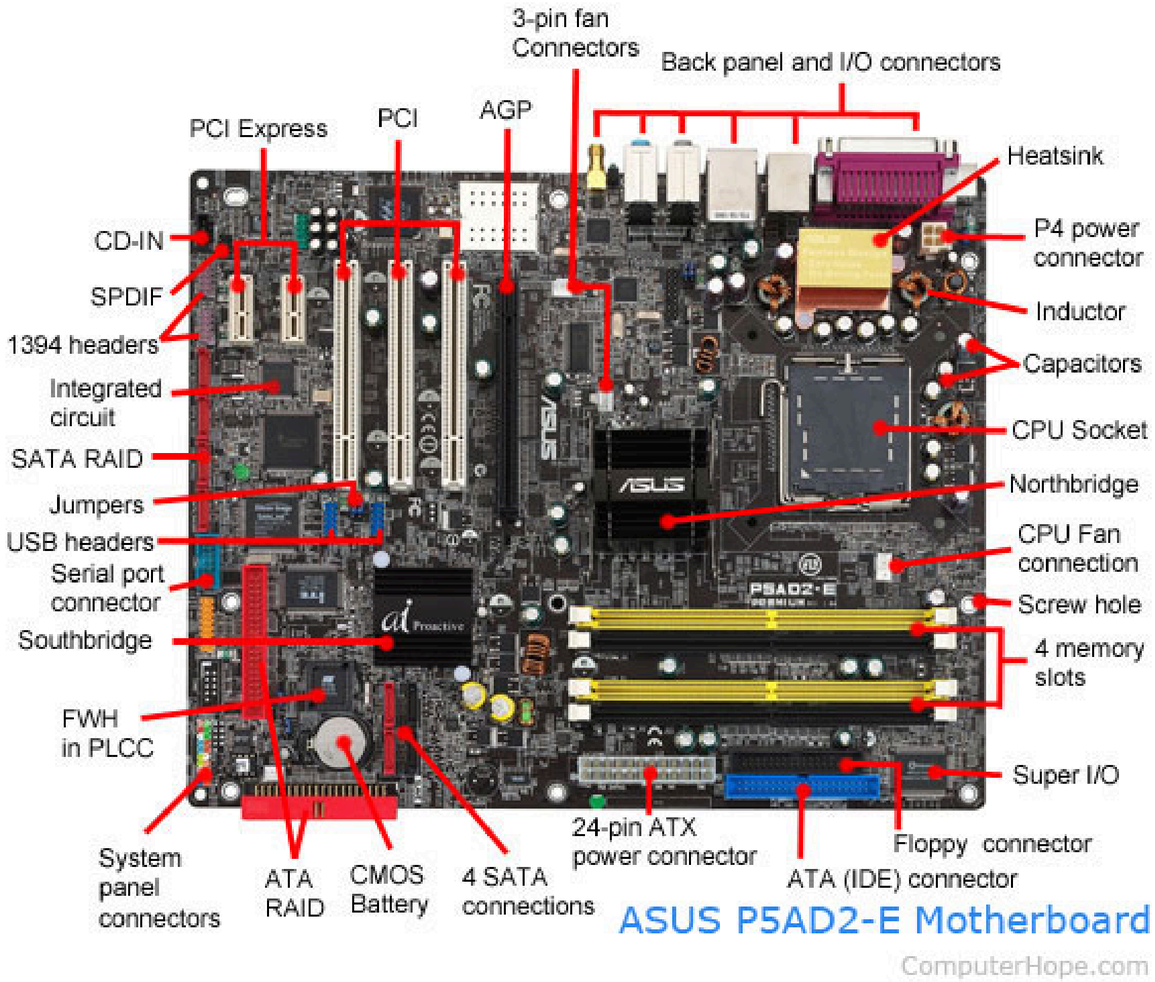


von Neumann architecture (1945)

A More Detailed Overview

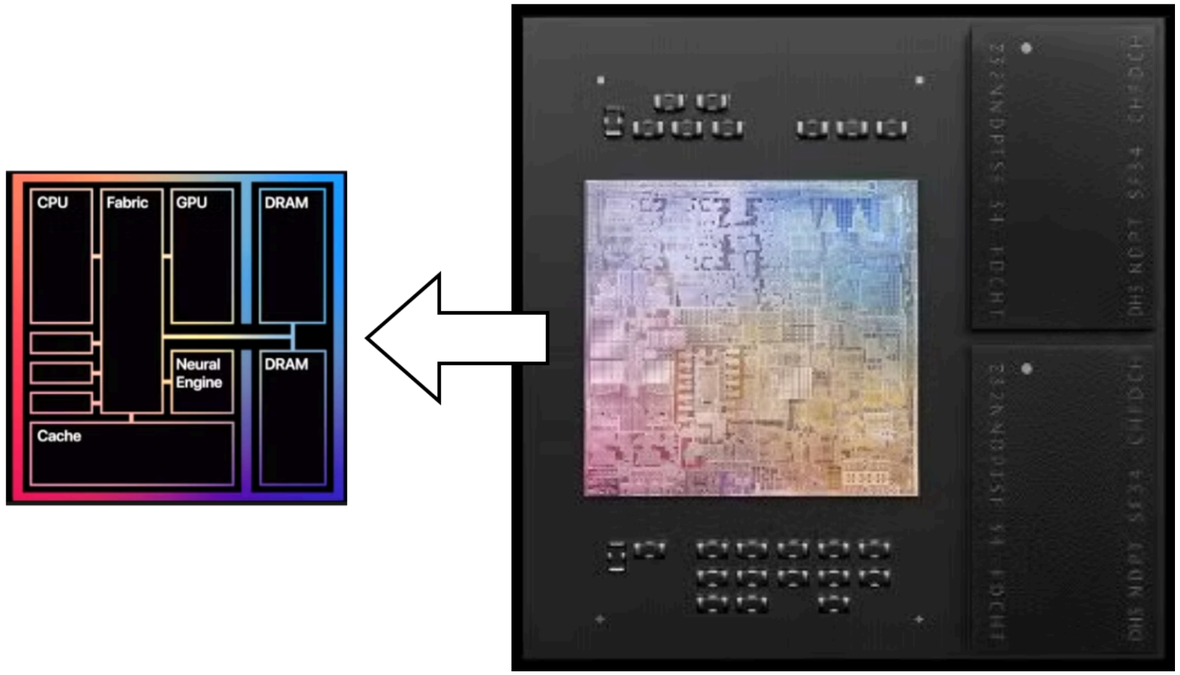


Motherboard

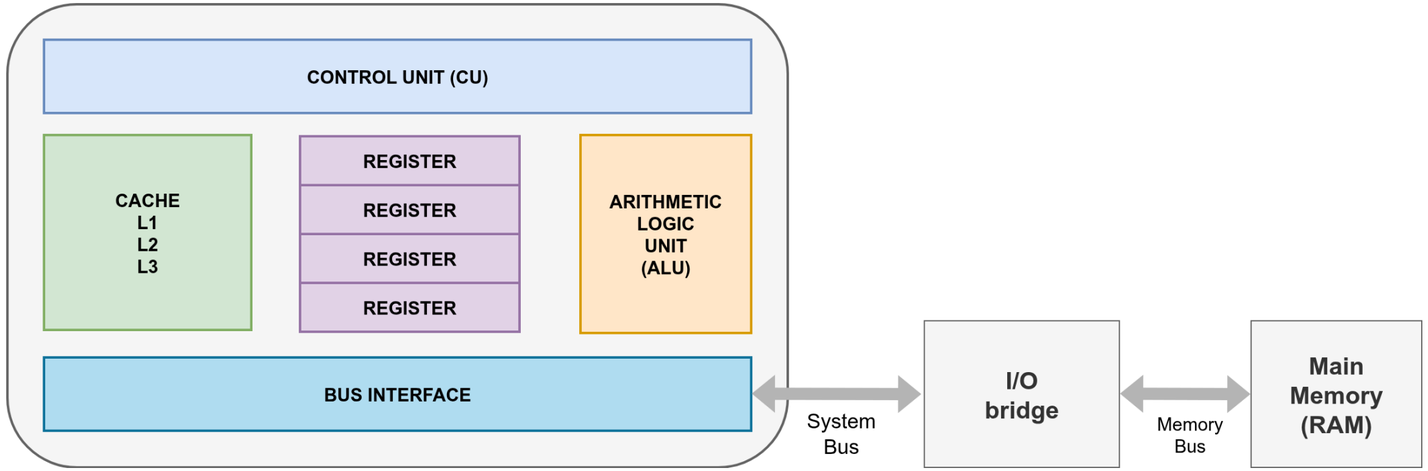


Modern System On Chip (SoC)

Modern SoCs integrate the CPU, GPU, storage, and other components on a single chip:



CPU (Central Processing Unit)



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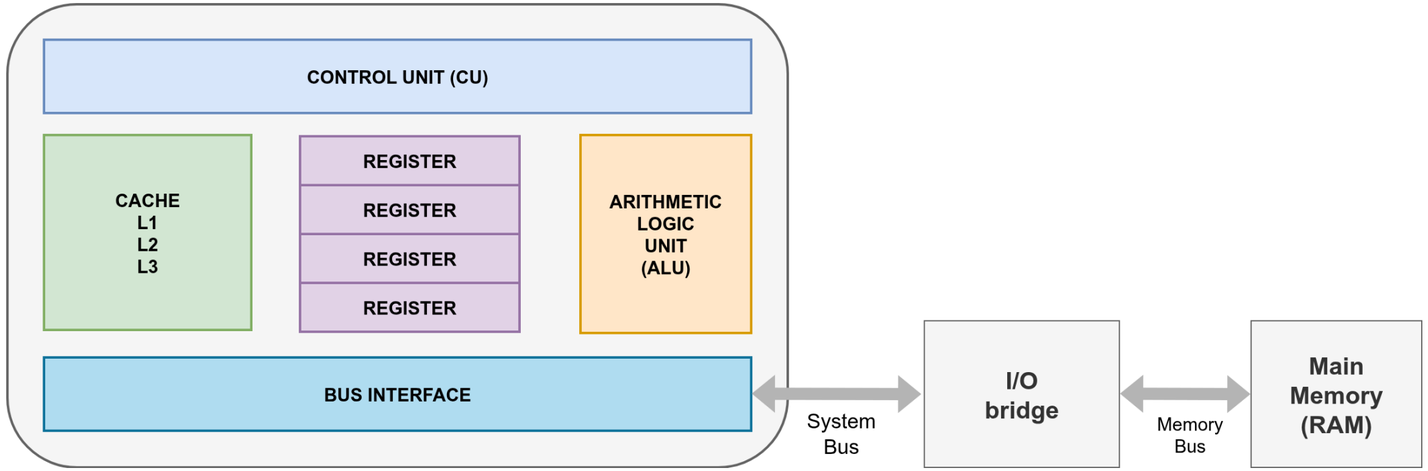

- The **Arithmetic Logic Unit (ALU)** performs arithmetic operations (such as addition and subtraction) and logical operations (like AND, OR, NOT) on binary data.
- The **Control Unit** coordinates the operations of the CPU, managing the flow of data and instructions (e.g., determining the next instruction to execute, performing the fetch-decode-execute cycle).
- The **Registers** are a few small (up to 64 bits wide), fast (<0.5 nanosecond) volatile memory locations that store data and instructions for processing.
- The **Cache** is a high-speed (1-16 nanoseconds) but limited (e.g., 1-16 MB) volatile memory that stores frequently used data to speed up access.

RAM (Random Access Memory)

RAM (Random Access Memory)

- The main (volatile) memory of the computer, where data and instructions are stored for processing before being loaded into the CPU.
- Implemented with Dynamic Random Access Memory (DRAM) chips, based on capacitors (require regular refreshes to maintain data).
- The capacity of RAM is nowadays on the order of gigabytes (GB) or terabytes (TB).
- Much faster ($1000 - 100000\times$) than secondary storage (e.g., hard drives, SSDs) but significantly slower ($10 - 100\times$) than CPU registers.
- Like a huge array of consecutive bytes, each identified by its address (e.g., $0, 1, 2, \ldots, 2^{64} - 1$).

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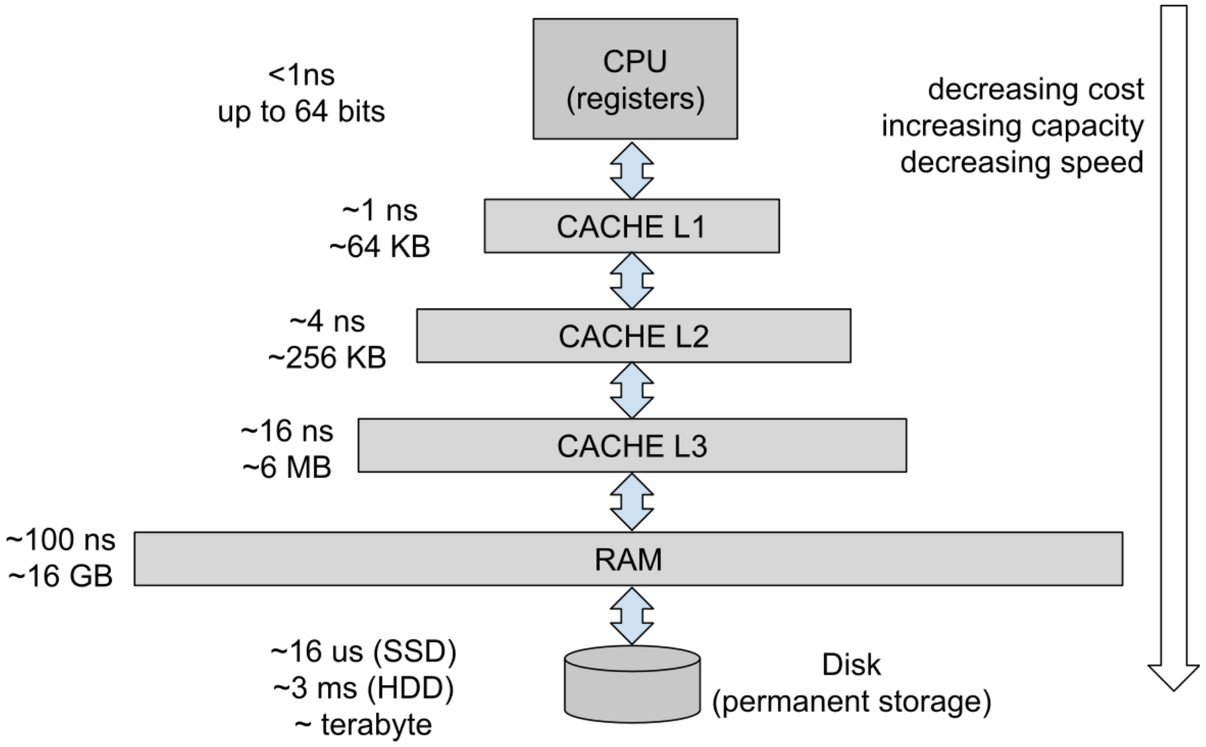
Disks can be connected to the computer in different ways:

- USB (Universal Serial Bus): used for external disks.
- SATA (Serial Advanced Technology Attachment): used for mechanical disks and older SSDs.
- NVMe (Non-Volatile Memory Express): used for modern SSDs.

Memory Hierarchy



Fast memory is expensive, hence a complex hierarchy is used to compromise between cost and performance: larger data is stored in slower, cheaper memory, that, when needed, is loaded into faster, more expensive memory in smaller amounts.

Software

How can a Computer Execute a Program?

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The machine code is:

- **Low level**: relatively simple operations
- **CPU specific**: different CPUs have different machine codes
- **Binary format**: humans struggle to read it

Machine Code: Intel/AMD vs ARM

A simple sequence of instructions to compute the sum of two integer numbers would look like this in machine code:

Intel/AMD 64 bit machine code

```
f3 0f 1e fa
55
48 89 e5
89 7d fc
89 75 f8
8b 55 fc
8b 45 f8
01 d0
5d
c3
```

ARM 64-bit machine code

```
ff 43 00 d1
e0 0f 00 b9
e8 0f 40 b9
e9 0f 40 b9
00 7d 09 1b
ff 43 00 91
c0 03 5f d6
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- Machine code is binary (shown here in hexadecimal for readability)
- Intel/AMD processors share the same machine code format
- Other processors like ARM use entirely different machine code formats

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How can programmers survive?

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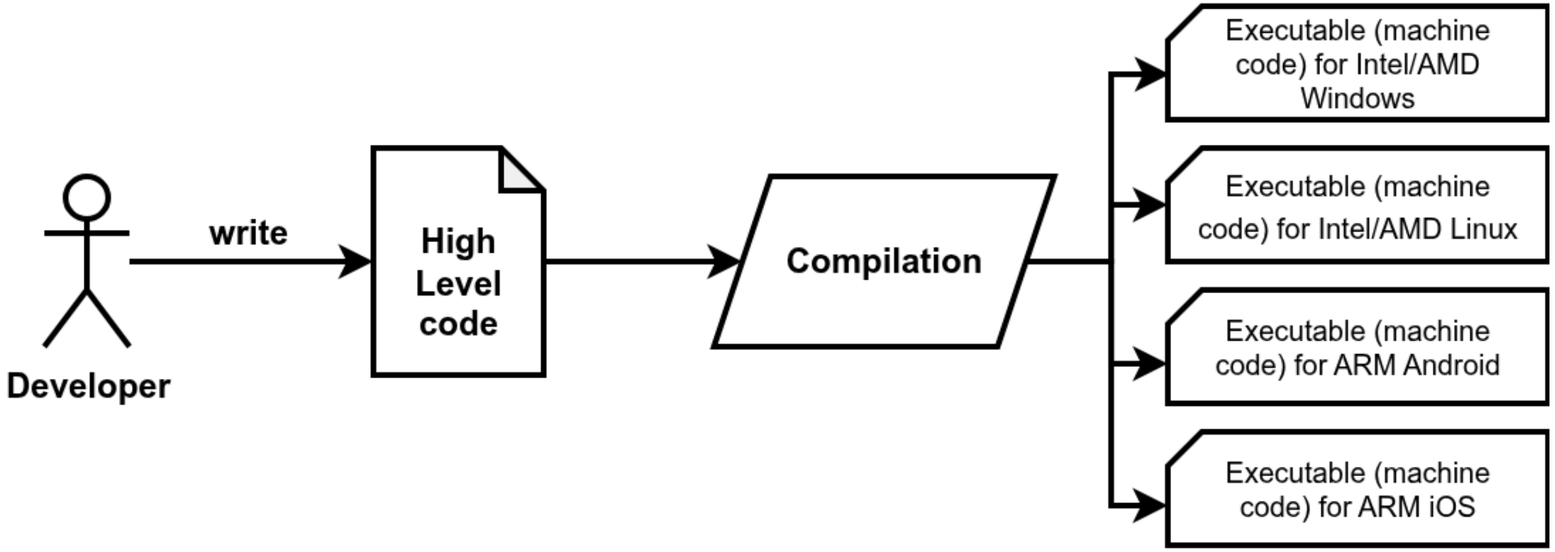
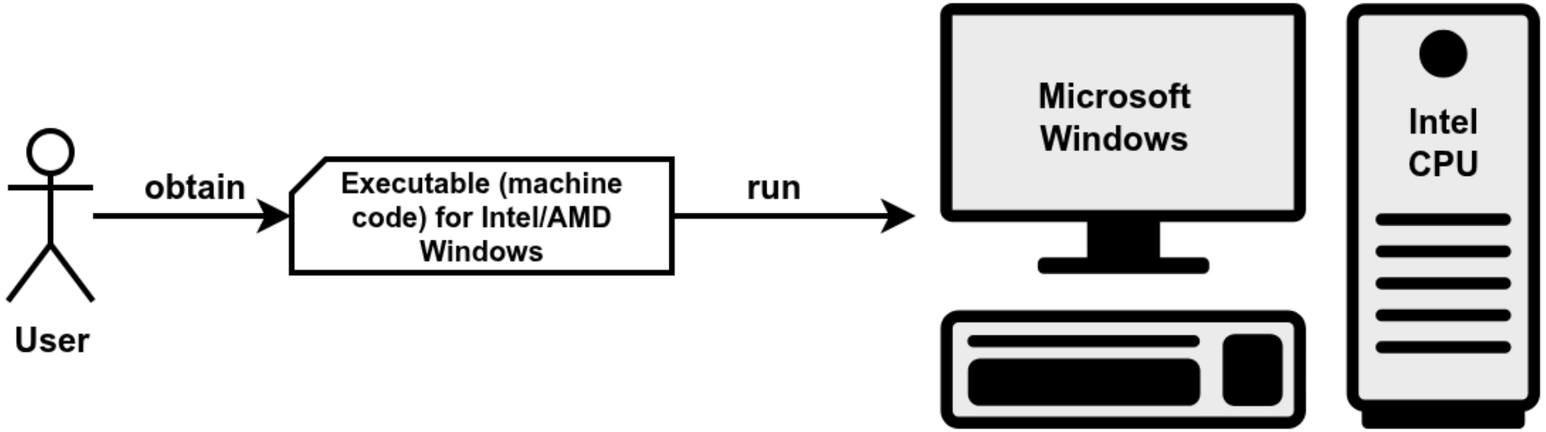
Code written in high-level languages must be translated into machine code. Two main strategies:

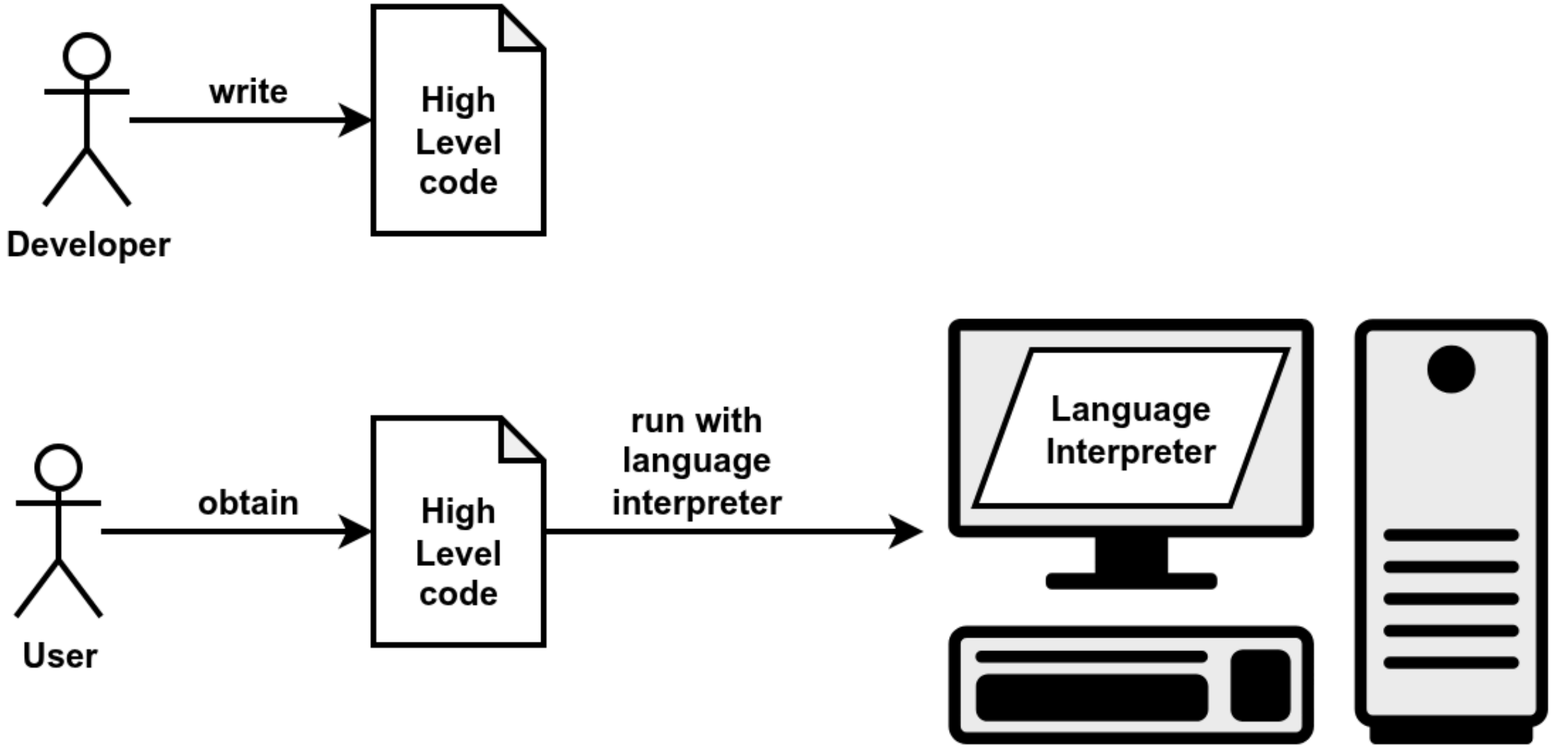
1. **Compilation**: compilers translate high-level code into machine code at *compile time* (before it is run). One executable for each platform.
2. **Interpretation**: interpreters translate high-level code into machine code at *running time*. The code is portable and platform independent.

Compilation





Interpretation



MANY Different High-Level Languages

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There are thousands of high-level languages:

- They have been designed with different goals in mind
- Some are general purpose, some are domain specific (e.g., web)
- Some gives more control to the user, some hide more details
- Some are compiled, some are interpreted

Computers Run Complex Software Stacks

Computers are expected to run **MANY** programs at the same time:

- Browser to visit web pages (e.g., Chrome)
- Office editor to write documents
- File manager to organize files
- Messaging applications (e.g., Telegram)
- Video streaming client (e.g., Netflix)
- ...and way more!

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This is a **LOT** of **HARD** work: we need an **Operating System** (**OS**).

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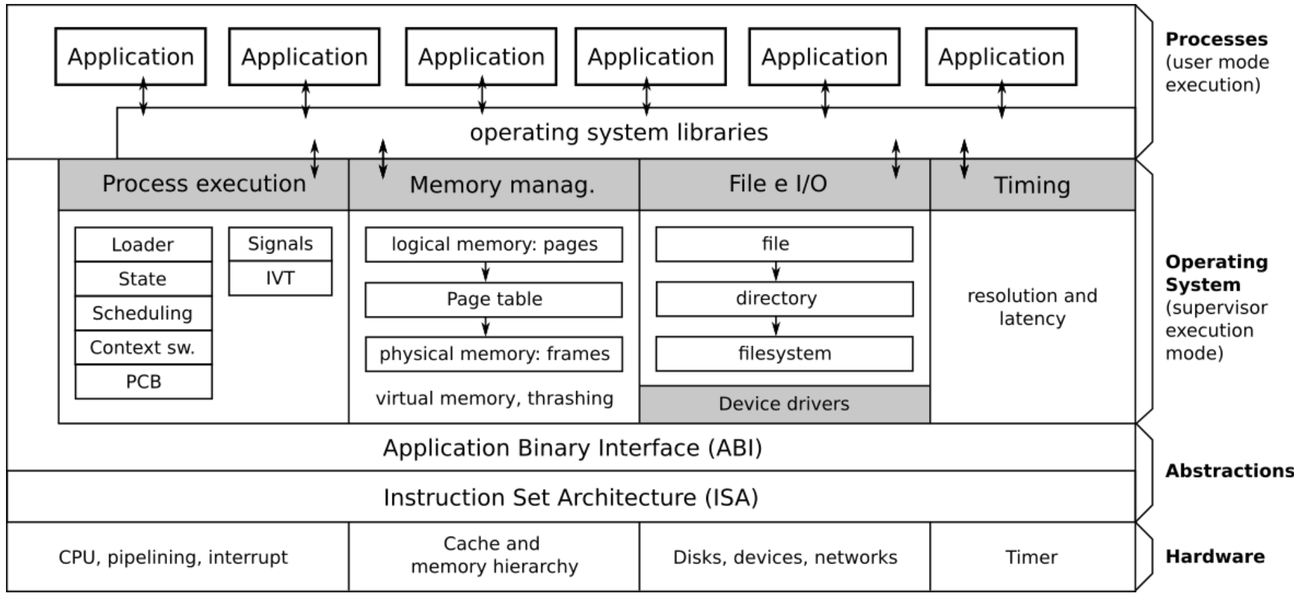
The main tasks of an operating system are:

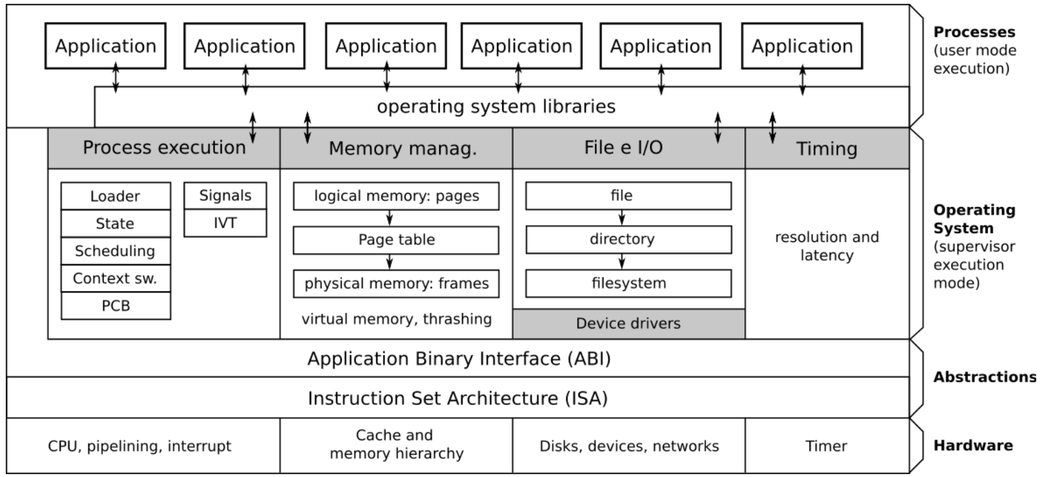
- **Process management** (creation, termination, CPU scheduling, etc.)
- **Resource allocation** to processes (RAM, storage, network connections)
- **User management** for multi-user systems (access privileges to files, programs)
- **Inter-process communication** (locally or across networked systems)
- **Error handling** during process execution or system operations
- **Protection and security**, isolating processes to prevent interference with other processes, the OS, or hardware

Overview of an Operating System



- The **hardware** is the physical part of the computer, containing CPU, RAM, disks and other devices.
- The **Instruction Set Architecture** (**ISA**) defines machine code instruction syntax and semantics.
- The **Application Binary Interface** (**ABI**) defines conventions on how different programs can interact, reuse code, and interface with the operating system.

Kernel Mode vs User Mode



The **operating system**:

- has full control over the hardware: it runs in **supervisor (kernel) execution mode**
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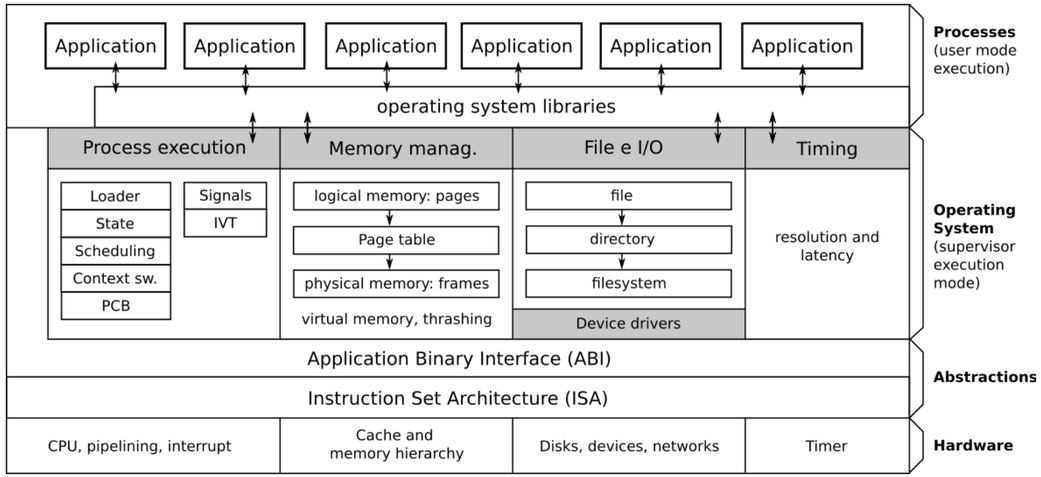

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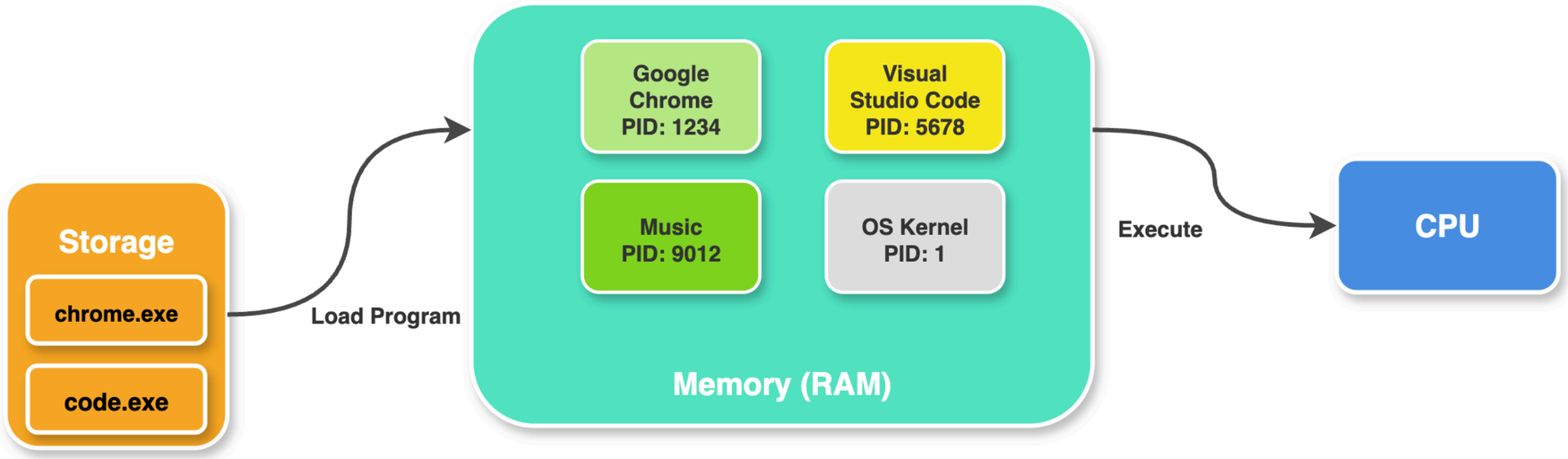
In **user execution mode**:

- Code can run only a limited (safe) subset of the ISA
- Privileged instructions must be delegated to the OS following the ABI
- Applications use system libraries to avoid reinventing the wheel

Processes

A **process** is a running program in memory:



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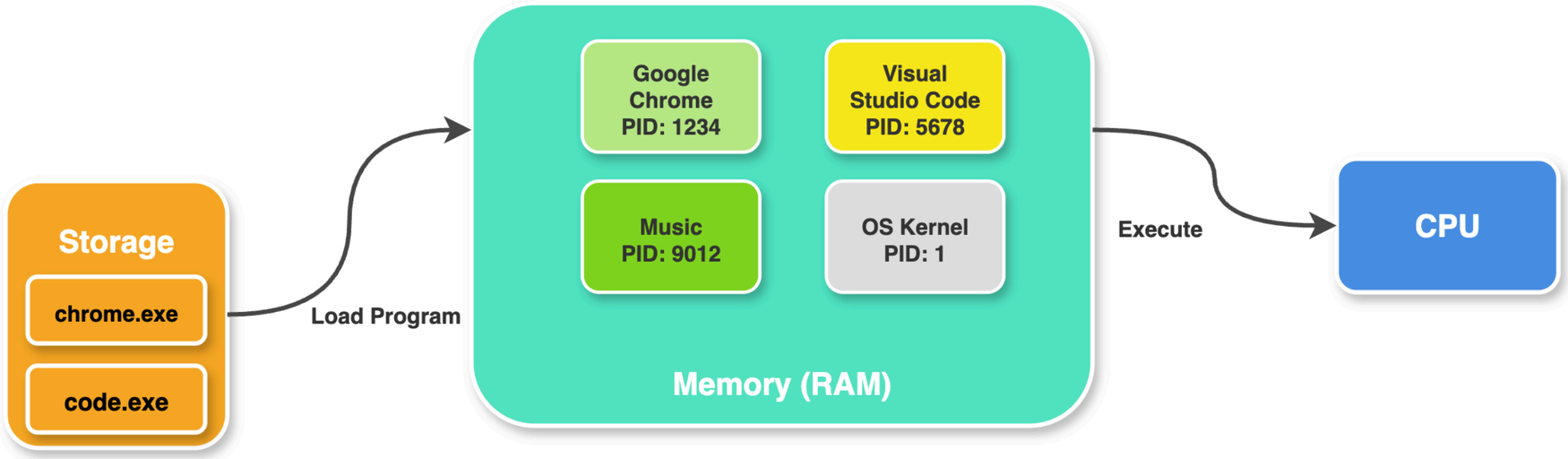

Key Process Concepts:

- **Process ID (PID)**: Unique identifier for each process
- **Parent/Child**: Processes can create other processes
- **Memory**: Each process has its own memory space
- **CPU Time**: Processes share CPU time

CPU Cores and Multitasking

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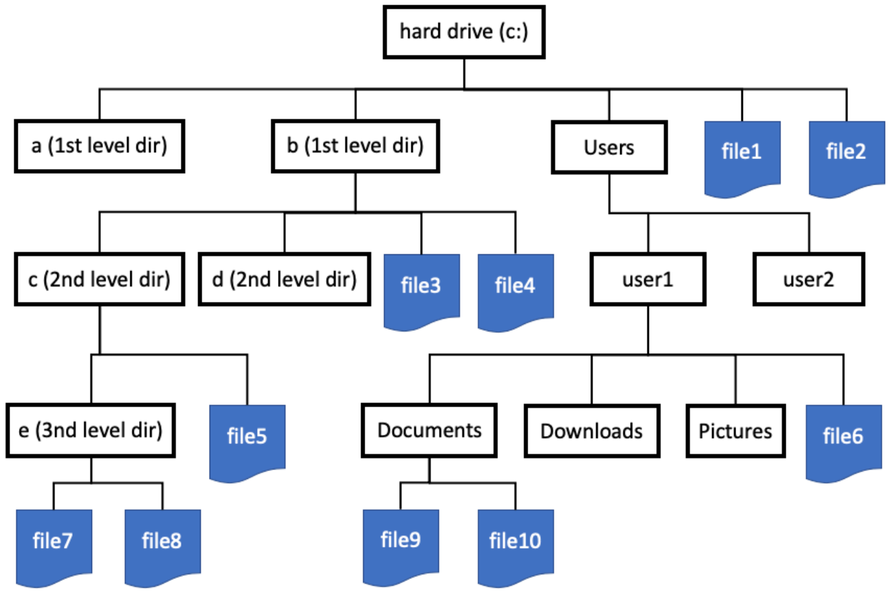
This technique is called **time-sharing** or **preemptive multitasking** and it is the essential trait of all modern operating systems.

File Systems

Every operating system organizes files in a **hierarchical structure**, like a tree:



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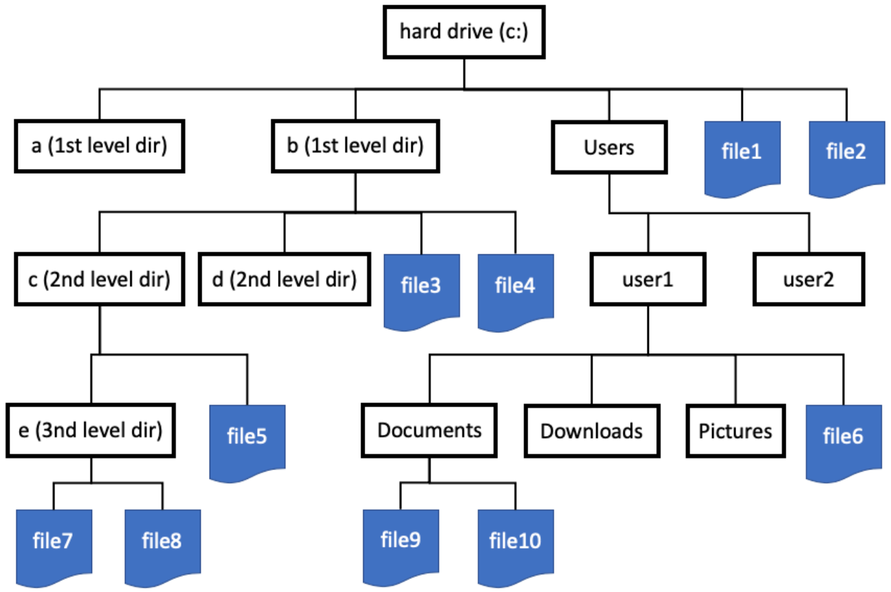

Key Concepts:

- **Root**: The top-level directory (starting point)
- **Directories/Folders**: Containers that hold files and other directories
- **Files**: Individual documents, programs, or data
- **Path**: The route to reach a specific file or directory

Application Installation Methods

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Each method has different **goals**, **advantages**, and **trade-offs** depending on:

- Security requirements
- Deployment complexity
- Resource constraints
- Maintenance needs

Virtual Machines

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Key motivations:

- **Isolation**: Applications run in their own environment without interfering with the host system
- **Compatibility**: Run software designed for a different OS
- **Security**: Containment of potential security vulnerabilities within the VM boundary
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A **Virtual Machine (VM)** is a software emulation of a physical computer that runs a complete operating system.

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- **Type 1 (Bare-metal)**: Runs directly on hardware (e.g., VMware ESXi, Microsoft Hyper-V, Xen)
  - Used in data centers and cloud environments
  - Better performance and security
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Each VM includes:

- Its own **guest operating system** (can be different from the host)
- **Virtual hardware** (virtual CPU, virtual RAM, virtual disk)
- **Applications** running inside the guest OS

Virtual Machines: Pros & Cons

Advantages:

- **Complete isolation**: Full OS-level separation between VMs
- **Run any OS**: Linux on Windows, Windows on macOS, etc.
- **Snapshots**: Save and restore the entire state of a VM
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Disadvantages:

- **Heavy resource usage**: Each VM runs a full OS (GBs of RAM, disk space)
- **Slow startup**: Minutes to boot a full OS
- **Performance overhead**: Hypervisor adds a layer of indirection
- **Portability**: VM images are large and hypervisor-dependent

Containers

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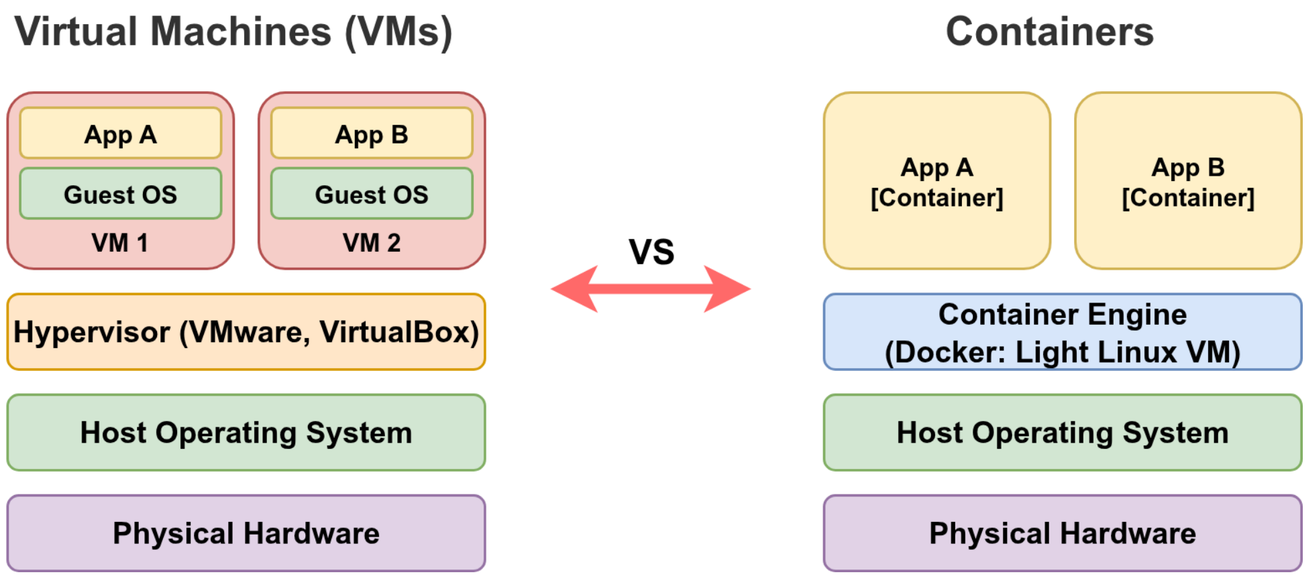
Unlike VMs, containers **share the host OS kernel** and isolate the application processes from each other using OS-level features (namespaces, cgroups).

Containers vs Virtual Machines



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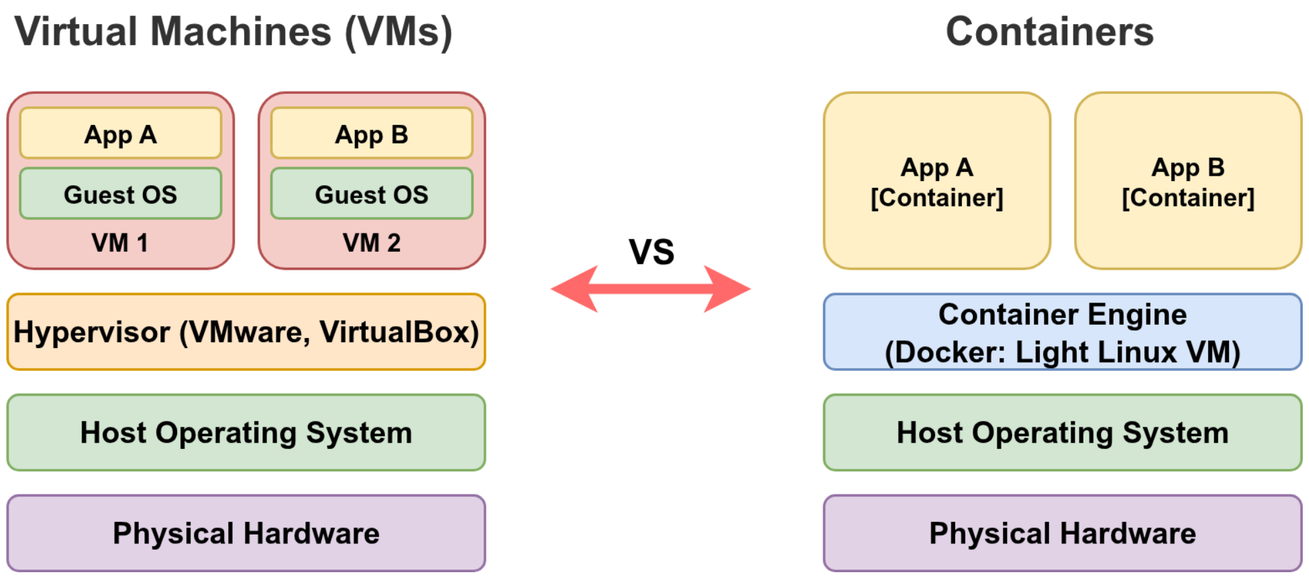

Aspect	Virtual Machines	Containers
Isolation	Complete OS isolation	Process-level isolation
Resource Usage	Heavy (full OS)	Lightweight (shared kernel)
Startup Time	Minutes	Seconds
Portability	Hypervisor dependent	Open format (OCI)

Containers: Pros & Cons

Advantages:

- **Lightweight**: Share the host OS kernel, minimal overhead
- **Fast startup**: Seconds instead of minutes
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Disadvantages:

- **Weaker isolation**: Shared kernel means less separation than VMs
- **Linux-centric**: On macOS and Windows, Docker requires a lightweight Linux VM
- **Not ideal for GUIs**: Primarily used for non-graphical applications
- **Security concerns**: A kernel vulnerability could affect all containers

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Containers have become the industry standard for deploying services in cloud environments, especially web services, due to their lightweight nature and efficient resource utilization.